Application of Explainable AI for Diagnosis of Alzheimer's: Layer-CAM based Analysis of Brain MRI scans

Kyu Won Park[1]*, Min Seong Kim[1]*, Moon Hwan Lee[2], Jae Youn Hwang[2]

DGIST, School of Undergraduate Studies[1],

DGIST, Department of Electrical Engineering and Computer Science[2]

*{kw040131, pushsoneg}@dgist.ac.kr (Equally contributed)

MBISLab Multimodal Biomedical Imaging and System

Introduction

Motivation and challenge:

- Alzheimer's Disease (AD) shows subtle and widespread early-stage changes, making early diagnosis difficult. While MRI detects key structural changes like hippocampal atrophy, the complexity of data limits conventional analysis. Deep learning offers solutions but suffers from a lack of clinical interpretability.

Solution and objective:

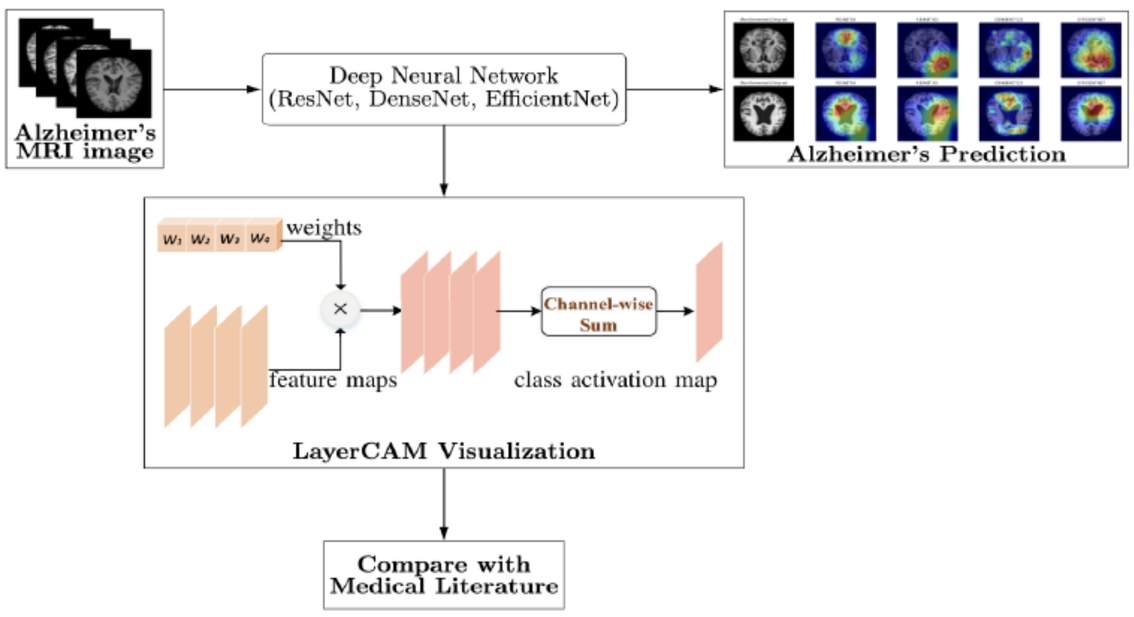
- We apply Layer-CAM to capture multi-scale features and validate clinical relevance in AD diagnosis. Four deep learning models (ResNet50, ResNet101, DenseNet121, EfficientNet) are trained and compared on the same dataset. Our aim is to identify models that highlight key pathological features, supporting more reliable early diagnosis of AD.

	'0' : Mild	'1' : Moderate	'2' : Non	'3': Very mild
Total	724	49	2566	1781
Train	559	37	2071	1429
Validation	84	6	249	173
Test	81	6	246	179

[Table 1. Class Distribution of the AD MRI Dataset]

Methods

- Trained deep learning models on 5,120 MRI scans labeled by Alzheimer's progression stage.
- Used Hugging Face dataset; split into training, validation, and test sets (8:1:1).
- Evaluated performance using accuracy, F1-score, precision, recall, specificity, and NPV.
- Applied LayerCAM to visualize attention maps.
- Verified model focus on clinically relevant regions (e.g., hippocampus, cortex).
- Explored clinical potential of explainable AI in Alzheimer's diagnosis.



[Figure 1. Overview of the Research Workflow]

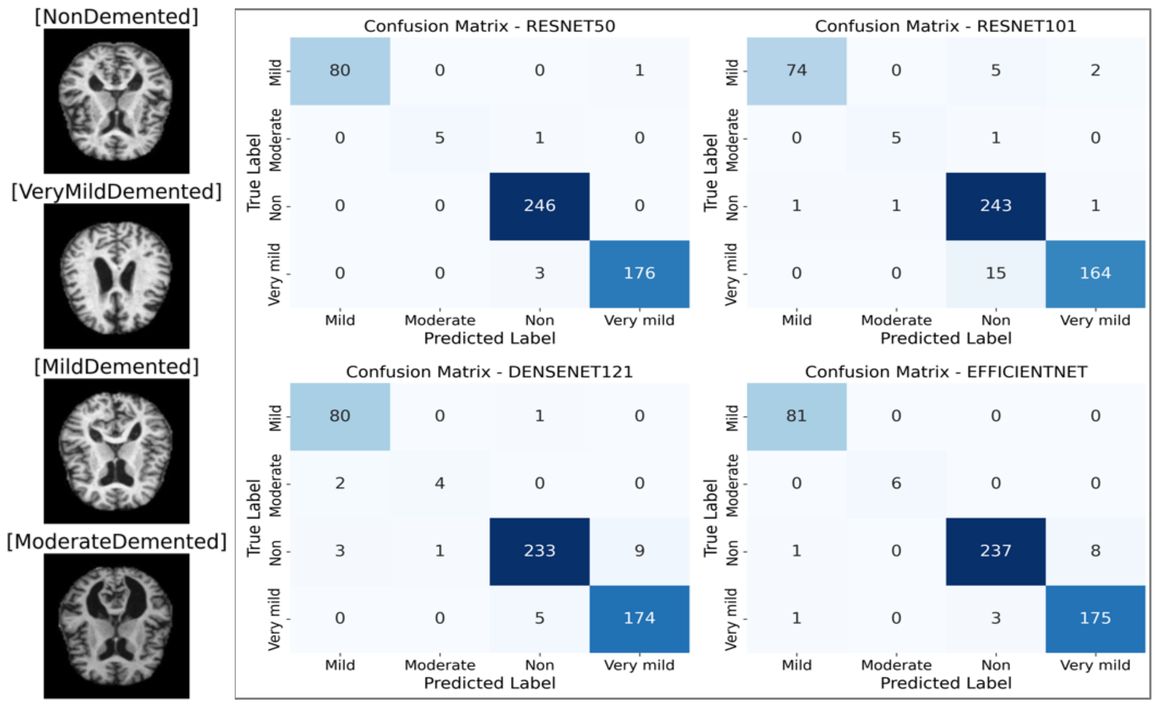
Result

A. Evaluation of Classification Accuracy

- Evaluated model performance for early Alzheimer's diagnosis using MRI.
- ResNet50 achieved the highest performance (>99% across all metrics), followed by EfficientNet, All models showed >90% accuracy, supporting the feasibility of MRI-based early AD detection.
- Confusion matrices indicated strong class discrimination and early detection capability.

Model	Accuracy (%)	F1 Score (%)	Precision (%)	Recall (%)	Specificity (%)	NPV (%)
ResNet50	99.02	99.01	99.03	99.02	99.55	99.67
ResNet101	94.92	94.91	95.14	94.92	97.69	98.16
DenseNet121	95.90	95.87	95.91	95.90	98.42	98.27
EfficientNet	97.46	97.46	97.49	97.46	99.00	98.87

[Table 2. Performance Comparison of Alzheimer's MRI Models]

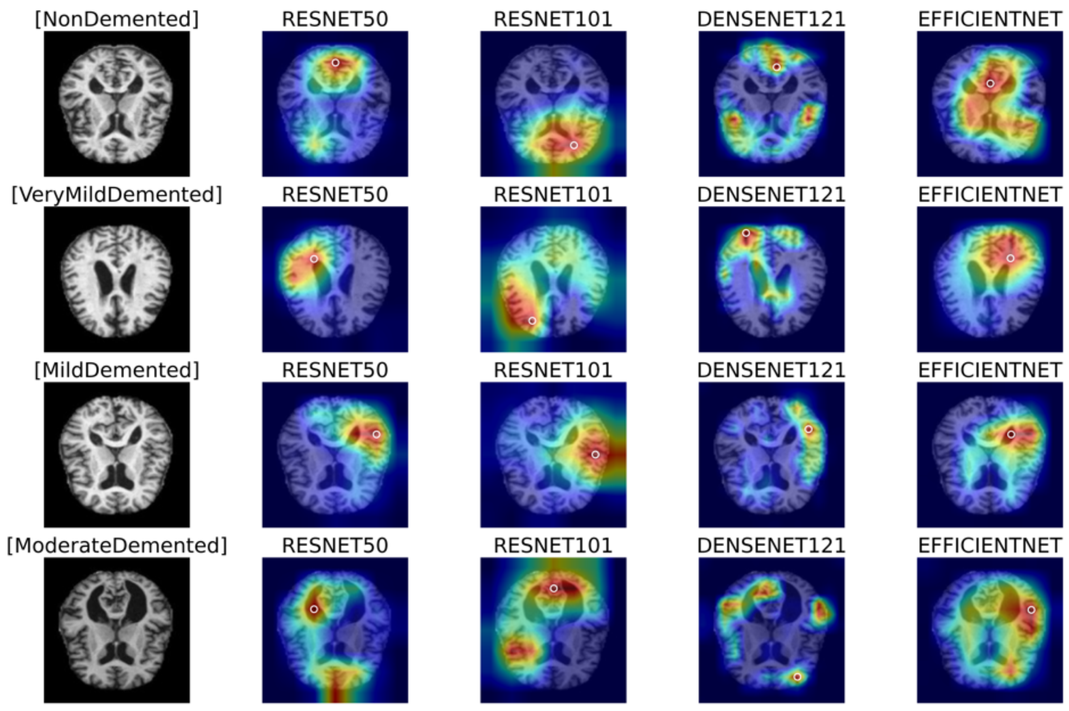


[Table 3. Confusion Matrix Results by Pathological Stage and Network]

B. CAM-based Explainability Analysis

- ***ResNet50***: Exhibited highly localized and consistent attention in the hippocampus and temporal lobe. Aligned well with established Alzheimer's-related biomarkers, particularly in Mild and Moderate Demented cases.
- ***EfficientNet***: Showed clinically meaningful focus, though with slightly reduced consistency compared to ResNet50.
- ***ResNet101***: Attention maps were inconsistent and scattered, limiting clinical interpretability.
- ***DenseNet121***: Produced diffuse and non-specific activation, occasionally highlighting irrelevant brain regions.

These results emphasize that explainability must accompany accuracy to ensure the clinical reliability of AI-based diagnostic models.



[Figure 2. Class Activation Map (CAM) Results]

Conclusion

- This study highlighted that classification performance alone is insufficient; **explainability is crucial for clinical AI applications**. While LayerCAM provided valuable insights, the lack of ground-truth lesion annotations limited quantitative validation of attention regions.
- ***Future Work***: Future studies will assess the reproducibility of attention patterns and cross-validate with PET imaging to confirm alignment with amyloid-beta deposition. Expert radiologist feedback will further validate the clinical relevance of highlighted regions. These efforts aim to enhance diagnostic performance and clinical trust, advancing precision medicine-based AI diagnostic systems.

Acknowledgements

이 연구는 산업통산자원부 및 산업기술평가원, 과학기술정보통신부 및 한국연구재단 지원에 의한 연구임. (RS-2024-00419246, RS-2023-00266120)

DGIST EECS ELECTRICAL ENGINEERING & COMPUTER SCIENCE

DGIST 대구경북과학기술원 Daegu Gyeongbuk Institute of Science & Technology